

Loan Sanction Amount Estimation

CS5100

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1. Introduction

Problem statement and motivation

Someone applying for a loan asks for a certain figure, and the bank almost never sanctions exactly that figure. Part of the request is usually cut, and a fair share of applications are refused outright and receive nothing at all. The applicant normally finds this out only at the end of the process, after weeks of paperwork.

This project estimates that figure early, from details an applicant already knows about themselves, such as income, credit score, existing loan expenses and the price of the property. It is not a lending decision and does not replace the checks a bank carries out, it is an early estimate so that a person can see roughly where they stand before committing time to an application.

Two lines of published work make the attempt worthwhile. Khandani et al. [1] show that machine learning models built on applicant transaction and credit bureau data forecast consumer delinquency and default considerably better than traditional scoring methods, so data of this kind carries enough signal to be modelled directly. Lessmann et al. [2] benchmark 41 classifiers across eight real credit scoring datasets and find that ensembles beat simpler baselines such as logistic regression fairly consistently. That second result shapes the method used here: several classifiers are tuned and compared on one fixed split rather than one being chosen in advance.

Goals

1. A model that takes a single application and predicts the sanctioned amount in US dollars, together with whether any amount is sanctioned at all.
2. An honest score for it, measured on applications it never saw during training and reported end to end rather than on whichever subset flatters each stage.
3. A record of what the raw file contained and which parts of it were unusable.

2. Background

The two references above set the terms for this work: [1] is the case that richer applicant data plus machine learning improves on scorecard style methods, and [2] is the case that the choice of classifier matters enough to be worth comparing several on one split instead of defending a favourite.

The decision to try tree ensembles before a neural network is not a matter of taste. Grinsztajn et al. [3] benchmark both families across many medium-sized tabular datasets and find that tree-based ensembles still come out ahead of deep networks on data of this shape. This dataset is tabular with 19 input columns and under 30,000 rows, so the tree ensembles were tuned first and the network was added afterwards as the comparison rather than the favourite.

Cragg [4] sets out a two-part model in which whether an outcome is zero is estimated separately from how large it is when positive. The target column here behaves exactly that way, because a sanctioned amount of 0 records a refusal rather than a very small loan. For interpretation, Lundberg and Lee [5] give a single framework for attributing a prediction to the features that produced it, with the attributions summing back to the prediction, which is what makes the SHAP values in Section 4.5 readable as shares of a total. Fuster et al. [6] find that moving credit underwriting onto machine

learning changes outcomes across demographic groups and in some cases widens existing gaps, and **Gender** is one of the features used here, which is why fairness appears as an open limitation in Section 5.

Stage 1 is therefore a binary classifier, tried as logistic regression, a random forest and a gradient boosting classifier, against a baseline that approves everyone, since 73.2 percent of this data is approved. Stage 2 is a regressor fitted only on approved applications, tried as linear regression, a principal component version of it, two tree ensembles and a PyTorch network, against a mean predictor.

3. Methodology

3.1 Tools and the dataset

The work runs on a Python 3.14.4 kernel, with pandas 2.3.3 and numpy 2.4.6 for data handling and scikit-learn 1.9.0 for every model except the network, so that preprocessing, grid search and metrics all come from one library. The network is written in PyTorch 2.9.1, because no TensorFlow wheel is published for Python 3.14.4. Attribution uses shap 0.52.0 and plots matplotlib 3.11.0 and seaborn 0.13.2.

The data is the Kaggle dataset “Predict Loan Amount Data”, available at <https://www.kaggle.com/datasets/phileinsophos/predict-loan-amount-data>. The file used is `train.csv`, which holds 30,000 applications in 23 columns once **Customer ID** is set as the index, keeping the identifier out of the feature set. The column being predicted is **Loan Sanction Amount (USD)**.

3.2 Data quality findings

A numeric sentinel was hiding inside several columns. -999 is used in this file to mean “not recorded”, and left in place it reads as a real quantity, so an unrecorded income would be averaged in with genuine incomes and drag the column down. Every -999 was converted to a blank before anything was counted. **Property Price**, **Co-Applicant** and **Current Loan Expenses (USD)** look almost complete on a raw null count, yet 352, 168 and 177 of their entries respectively were the sentinel rather than an empty cell.

678 rows have no target and were dropped, leaving 29,322. A missing target cannot be filled, because filling it would mean inventing the answer and training the model to reproduce the invention. Of those 678 rows, 338 carried -999 rather than a blank.

One column is an exact duplicate of another. **Property Age** is not an age. It is a copy of **Income (USD)**, identical to the cent in all 25,150 rows where both values are present, giving a correlation of exactly 1.000000. A match that exact only happens when one column was copied from the other, so **Property Age** was dropped, along with **Name** and **Property ID**, which are nearly unique per row and can only be memorised. That leaves 20 columns, 19 of them inputs.

Over a quarter of the target column is not an amount. 7,865 applications, 26.8 percent, have a sanctioned amount of exactly 0, against 21,457 that receive something. A 0 here records a refusal, not a very small loan, and refusals are not spread at random. An applicant with high income stability is refused 14.8 percent of the time against 29.0 percent for low income stability, and 30.1 percent of rural applicants are refused against 21.8 percent of urban ones. Refusal is

predictable from fields available at application time, so those rows carry a pattern worth learning rather than noise worth discarding.

Type of Employment is absent in 24.30 percent of rows and **Income (USD)** in 15.04 percent. Both are too large to drop, so both are imputed.

3.3 Splitting before looking

The split was made before any chart was drawn, because if the rows used for grading are looked at first then the test set stops measuring generalisation. 23,457 rows go to training and 5,865 to the held-out test set, an 80/20 split of the 29,322 remaining, stratified on approval status so that the approval rate lands at 0.732 in both halves.

3.4 The preprocessing pipeline

Every transformer sits inside a scikit-learn **Pipeline**, so that under cross validation it is refit on each training fold and a held-out fold is transformed using statistics it did not contribute to. The 19 input columns fall into the three groups shown in Figure 1.

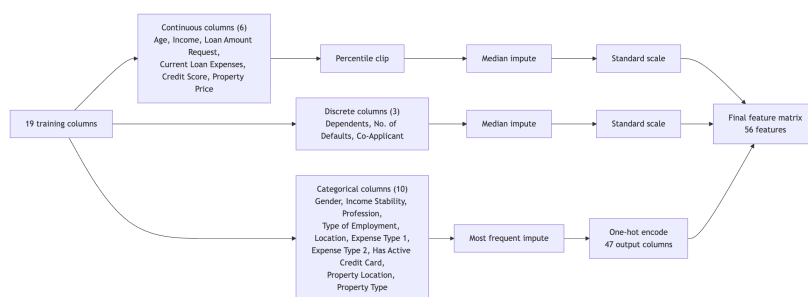


Figure 1: The preprocessing pipeline, 19 columns to 56 features

Two choices there are worth the reason behind them. Nominal categories are one-hot encoded rather than numbered, because numbering **Profession** as 0, 1, 2 and 3 would tell a linear model that category 3 is three times category 1, which is not true of a job title. **Property Type** is treated the same way despite being stored as an integer, since its values are codes rather than a count. Outliers in the six continuous columns are clipped at the 1st and 99th percentiles rather than replaced by the column mean, because clipping keeps every row in rank order and invents nothing. The three discrete count columns skip clipping, because fourteen dependents is an unusual household rather than a typo.

The pipeline produces 56 features from 19 columns. Refitting it on approved rows only, as Stage 2 requires, gives 54, because the **Profession** values “Unemployed” and “Student” appear on only 2 training rows and both were refused.

3.5 The two stage design and the metrics

A single regressor fitted over the whole target column has to aim at 0 for a quarter of the rows and at a five or six figure sum for the rest, so it settles in between and serves neither group. Following the two-part structure in [4], the work splits as shown in Figure 2.

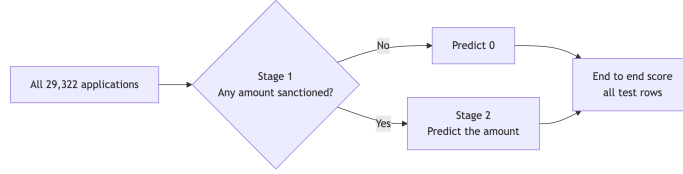


Figure 2: The two stage design

Stage 1 predicts whether anything is sanctioned and Stage 2, trained only on approved applications, predicts how much. An application that Stage 1 marks as refused is reported as 0 and never reaches the regressor. The refused rows are not deleted from the project, because for those applicants 0 is the correct answer and a model that has never seen a rejection has no way of saying no.

The headline regression metric is RMSE in US dollars, so the error reads as money, with R2 beside it because a low RMSE on a target that barely varies is not an achievement. Stage 1 is judged on F1 rather than accuracy, since a model that approves everyone already scores 0.7318 accuracy here, so accuracy on a split this uneven mostly measures the split.

4. Results

4.1 Stage 1, the approval classifier

All four are scored on the 5,865 held-out test rows.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline, always approve	0.7318	0.7318	1.0000	0.8451	0.5000
Logistic regression	0.8805	0.8697	0.9842	0.9234	0.8582
Random forest	0.9158	0.8971	0.9995	0.9456	0.8866
Gradient boosting	0.9168	0.8982	0.9995	0.9462	0.8894

All three tuned models clear the baseline F1 of 0.8451 by a wide margin, and gradient boosting leads on accuracy, precision, F1 and ROC-AUC together, so it was kept. Recall of 0.9995 is where the result starts to look less impressive than the F1 suggests, since approved rows are the easier half of this target and there are nearly three times as many of them. Precision, at 0.8982, is where the harder half shows up.

Figure 3 makes the imbalance concrete. The model produces 486 false approvals against 2 false rejections. Set against the 1,573 applications actually refused in the test set, roughly 3 in 10 refused applicants are wrongly told a number is coming, while almost no viable applicant is turned away, at 2 out of 4,292. The model has learned to say yes, an asymmetry invisible in an F1 of 0.9462 that drives the rest of the results.

Figure 4 plots the ROC curve for each tuned classifier, with a marked point at the default 0.5 threshold where each model actually operates. Random forest and gradient boosting land at 0.8866 and 0.8894 AUC, only 0.0028 apart, which is why each gets its own panel rather than being read off one shared curve. On every curve the 0.5 point sits high, close to a catch rate of 1, so the curve above that point is mostly trading further recall for false alarms, which is the same 486 to

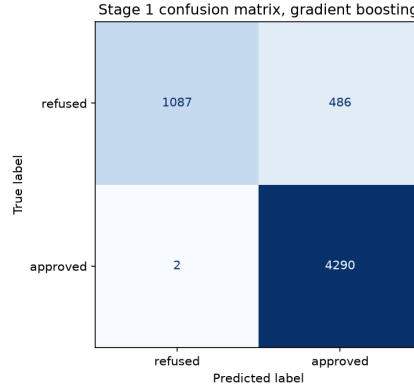


Figure 3: Stage 1 confusion matrix on the held-out test set

2 imbalance the confusion matrix already reports. The always-approve baseline has an AUC of exactly 0.5000, so it is not drawn separately, it is the diagonal chance line already shown in each panel.

4.2 Stage 2, the amount regressor

Six models are scored on the 4,292 approved test rows.

Model	RMSE (USD)	MAE (USD)	R2
Dummy, mean of training target	44,861.14	35,106.15	-0.0000
Linear regression	6,339.64	3,925.84	0.9800
PCA to 10 components, then linear regression	11,428.85	8,066.47	0.9351
Random forest	5,208.24	3,230.15	0.9865
Gradient boosting	5,137.77	3,143.83	0.9869
Neural network, PyTorch MLP	18,636.00	14,464.95	0.8274

Gradient boosting wins and becomes the Stage 2 model. Ten principal components hold 71.23 percent of the variance, which sounds acceptable until RMSE is seen to rise by 80 percent. The network finishes behind plain linear regression, consistent with its own training curve, where validation loss reaches its lowest point at epoch 25 and then climbs while training loss keeps falling, and with the pattern reported in [3] for tabular data of this size.

An R2 of 0.9869 is not the achievement it appears to be. A plain linear regression fitted on **Loan Amount Request (USD)** alone, with no other column and no tuning, reaches an RMSE of 6,123.84 USD and an R2 of 0.9814, which is not worse than the 54-feature linear regression above, it is slightly better. The random forest agrees, placing 98.05 percent of its feature importance on that one column. The sanctioned amount is anchored to the requested amount, at a Pearson correlation of 0.9907 among approved training rows and a median sanctioned share of exactly 0.700 of what was asked for. The high score here is a property of the data, not evidence that any model has learned how a bank underwrites a loan.

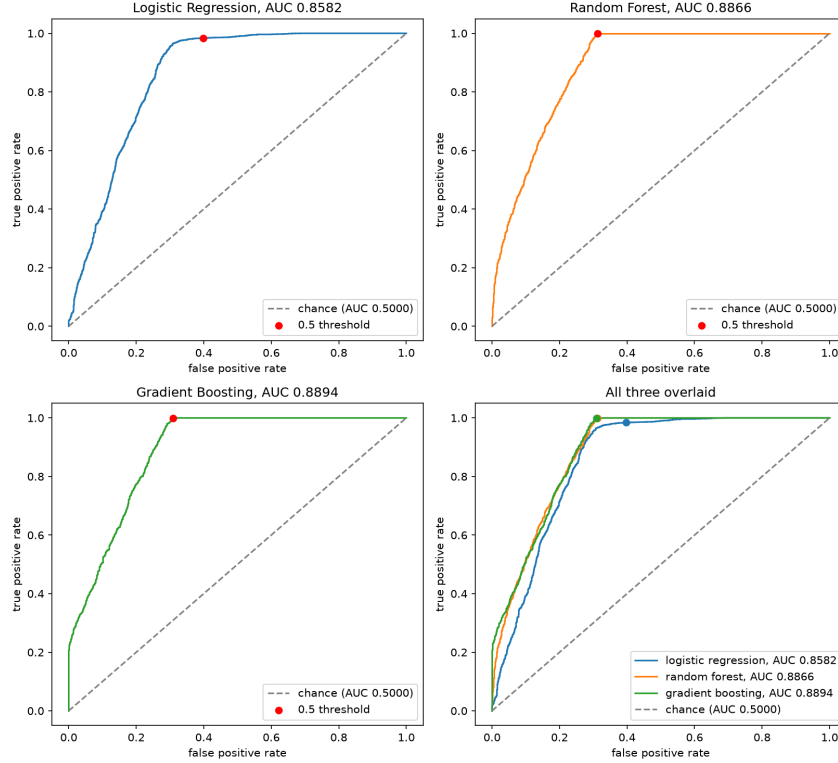


Figure 4: ROC curves for the three stage 1 classifiers, with the 0.5 operating point marked

4.3 End to end, the number an applicant would actually see

Stage 1 was scored on all test rows and Stage 2 only on genuinely approved rows, the set that excludes the mistake costing the most. Chaining the two pipelines gives the figure a real applicant would receive.

Setting	Rows scored	RMSE (USD)	MAE (USD)	R2
Predicted gate, Stage 1 decides	5,865	22,814.62	7,706.60	0.7757
Perfect gate, true label decides	5,865	4,395.12	2,300.65	0.9917

This is the honest number and it is much worse. RMSE rises from 5,137.77 USD to 22,814.62 USD and R2 falls from 0.9869 to 0.7757, which is not the same error measured over more rows but a genuinely worse one, because a wrong approval costs an applicant the entire predicted loan amount rather than a percentage of it. Replacing the model's gate with the true label brings RMSE down to 4,395.12 USD and lifts R2 to 0.9917, better than Stage 2 on its own subset, because a perfect gate assigns exactly 0 to every truly refused row and those 1,087 zero-error rows pull the average down. Almost all of the distance from 4,395.12 to 22,814.62 sits in Stage 1 choosing the wrong gate.

4.4 Where the error comes from

Splitting the squared error by what Stage 1 did with each row gives the sharpest result in the project.

Stage 1 outcome	Rows	Share of total squared error
Falsely approved	486	94.81%
Correctly approved	4,290	3.70%
Falsely rejected	2	1.50%
Correctly refused	1,087	0.00%

The 486 falsely approved rows are 8.3 percent of the test set, and the correctly refused rows contribute exactly nothing, because reporting 0 for an application that is truly refused is not a partial hit, it is exact. A handful of misclassified rows, and not the ordinary spread of regression error, decides almost the entire end to end score. That says precisely where any further effort should go.

4.5 What each stage relies on

SHAP values [5] were computed on 500 sampled test rows for each stage.

For Stage 1 the top three features are **Credit Score**, **Co-Applicant** and **Income (USD)**, together holding 74.5 percent of the total SHAP magnitude across 56 features, with mean absolute values of 1.3244, 0.9495 and 0.4657. A low credit score or the absence of a co-applicant pushes a prediction toward refusal, the direction a loan officer would expect. The attribution is spread across several columns, which is the sign that this stage is reading the application.

Stage 2 looks nothing like that. **Loan Amount Request (USD)** alone holds 90.0 percent of the total SHAP magnitude, rising to 98.7 percent once **Credit Score** and **Property Price** are added. The mean absolute SHAP value for the requested amount is 34,798.73 against 3,073.61 for credit score and 289.29 for property price, so the other two nudge the figure rather than shape it. This confirms the 98.05 percent feature importance of Section 4.2 from a second direction. Neither stage's attribution says a feature causes an outcome, since SHAP only attributes credit for the prediction this model makes.

5. Discussion

5.1 Interpretation

Stage 2 posts an R2 of 0.9869 and is the least interesting part of the project, because the sanctioned amount is anchored to the requested one and a single column reaches 0.9814 on its own. Stage 1 posts an F1 of 0.9462 and is where the difficulty sits, because that number hides 486 false approvals against 2 false rejections. The end to end RMSE of 22,814.62 USD is the figure that should be quoted, and 8.3 percent of rows carry 94.81 percent of it, every one of them a Stage 1 mistake. A genuine refusal is far more often reported to an applicant as an approval with a dollar figure attached than the reverse.

5.2 Limitations

The dataset itself is not fully trustworthy. It is a Kaggle extract with no documented provenance, and the audit found **Property Age** to be a column labelled and shipped as something it is not. One confirmed error of that kind, found only because it was checked, is reason to treat the rest of the file as unverified.

Stage 2 is close to a one-feature problem. This is set out in Section 4.2 and is the single most important caveat on the headline scores.

Stage 1 leans toward approval and nothing corrects for it. The decision threshold was left at the default 0.5 and no class weighting was applied, so the 486 to 2 asymmetry was accepted as it came out rather than tuned against.

The neural network overfits and nothing corrects for that either. Validation loss reaches its lowest point at epoch 25 and climbs through epoch 50 while training loss keeps falling, and the network is still scored at epoch 50, because no early stopping was built into the training loop.

No fairness audit was carried out, even though Gender is a feature of both stages. The target is a historical lending decision, so any bias in how those loans were sanctioned is reproduced rather than corrected. Given the findings in [6], this is a real gap.

The model reflects one snapshot of lending behaviour. It is fitted once on a fixed file, so if the conditions that produced these 29,322 applications shift, through interest rates, underwriting rules or the mix of applicants, nothing here would notice.

5.3 Improvements

The error decomposition sets the priority and it points at Stage 1 alone, since improving Stage 2 further would move 3.70 percent of the error at best.

The first change is to tune the Stage 1 decision threshold rather than leave it at 0.5. Precision of 0.8982 against recall of 0.9995 is a lopsided trade, and raising the threshold would convert some of that spare recall into precision. Because 94.81 percent of the error sits on falsely approved rows, even a small precision gain should move the end to end RMSE noticeably, and class weighting would push the same way.

The second is to score the two mistakes differently instead of treating them as equal, since F1 gives a false approval and a false rejection the same weight but they cost very different amounts. A cost-sensitive objective would rank Stage 1 candidates closer to the way the chained system is actually judged.

The third is a fairness audit across **Gender** and the other applicant attributes, comparing refusal rates and sanctioned amounts with the remaining columns held fixed. Adding early stopping to the network is worth doing as well, though on the evidence in [3] it would not overtake the tree ensembles here.

6. Conclusion

6.1 What was achieved

A two stage model predicts the loan amount an applicant is likely to have sanctioned. Stage 1 reaches an F1 of 0.9462 and Stage 2 an R2 of 0.9869 on held-out data, and chained together they give an end to end RMSE of 22,814.62 USD and an R2 of 0.7757 across all 5,865 test rows. The project also produced a data quality audit that found an exact duplicate column and a numeric sentinel, an error decomposition locating 94.81 percent of the error in 8.3 percent of the rows and a SHAP analysis showing that Stage 2 rests on one input column.

6.2 Lessons learned

A good score can be a property of the data rather than of the model. Stage 2's R2 of 0.9869 looked like the strongest result in the project until a one-column regression reached 0.9814, at which point the score turned into a description of how the target is constructed.

The scoring subset also decides the story, since each stage scored on its own rows is flattering and only the chained evaluation answers what an applicant would experience. An F1 of 0.9462 likewise gives no hint of a 486 to 2 imbalance, which only the confusion matrix and the error decomposition exposed.

6.3 Individual contributions

Member	Contribution
Kushal Krishnappa	Data quality audit and the preprocessing pipeline, including sentinel handling, the duplicate column finding, clipping and encoding
Deepak	Exploratory data analysis and all figures, from univariate through to correlation analysis
Rohit	Stage 1 classifier, the threshold and metric choice and the end to end evaluation
Dips	Stage 2 regressors including the PyTorch network, plus the SHAP interpretation

Report writing and final review were shared across all four members.

7. References

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